UNIT 11 HETEROSCEDASTICITY [*]

Structure

11.0 OBJECTIVES

After going through this unit, you should be in a position to

- explain the concept of heteroscedasticity in regression analysis;
- explain the consequences of heteroscedasticity;
- apply various methods used to detect heteroscedasticity; and
- provide solutions to the problem of heteroscedasticity.

11.1 INTRODUCTION

In the classical regression model that we discussed in Block 1 we assumed that the error terms are identically independently distributed (IID) with mean 0 and variance σ^2. An implication of the above is that the error variance (σ^2) remains constant for all observations. This assumption of constant variance, however, does not remain valid always. It may so happen that the errors are mutually uncorrelated (that is, no serial correlation) but have different variances. Thus, when the variance of the error term increases or decreases with the dependent variable, we have the case of heteroscedasticity. In other words, we say that the problem of heteroscedasticity arises when the assumption of homoscedasticity – that the variances of the stochastic disturbance term are finite and constant over the sample – is not met.

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[*] Dr. Priti Agrawal, O P Jindal Global University, Sonipat

Heteroscedasticity usually occurs in cross-section data. For example, consider the savings behaviour of households. We know that savings is relatively higher among households with higher income. It is also likely that the variance of savings among high-income families is larger than the variance among low-income families. Low-income families do not have much to save, so they cannot differ appreciably in their levels of savings. High-income families, by contrast, exhibit wide ranges, from the miser who saves virtually all his income to the spendthrift who saves virtually nothing. Secondly, there is more freedom of choice for high-income families while low-income families have to spend a major part of their income on consumption goods. Clearly, a cross-section sample of data on savings and income behaviour to estimate a regression model may not satisfy the homoscedasticity assumption.

There could be several other reasons for the presence of heteroscedasticity: There may be certain outliers in the data which would increase or decrease error variance. Further, heteroscedasticity often arises because the scale of a variable varies enormously within the sample. For example, in India the large states have relatively large state domestic product (SDP) compared small states. If we take SDP as a regressor in a model, then the variability is likely to increase enormously as we move from small to large states. However, if we take per capita SDP then the variation would not be so prominent. While Maharashtra's SDP is 175 times that of Manipur, Maharashtra's per capita SDP is about 2.5 times that of Manipur. The point we try to make is that heteroscedasticity in many cases could be due to incorrect data transformation. Thus, variables should be appropriately transformed before running a regression.

Heteroscedasticity, or unequal error variance, is more likely to be present in cross section data than in time series data. In the case of time series data, the same entity or unit is surveyed over a period of time which reduces the chances of variability in errors.

11.2 CONCEPT OF HETEROSCEDASTICITY

As pointed out above, error terms are said to be homoscedastic if the variance of the error term u_i conditional upon x_i is equal to σ^2.

$$E({u_i}^2) = Var(u_i) = \sigma^2 \qquad \ldots (11.1)$$

(i = 1, 2, ..., n)

In Fig.11.1 below, we show the homoscedasticity condition through a three-dimensional diagram. We observe that the conditional variance of the dependent variable the same for all the values of x_i (that is, x_1, x_2 and x_3).

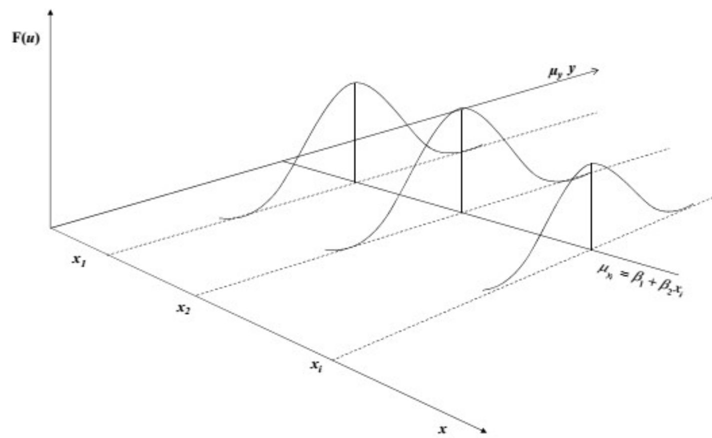

Fig. 11.1: Homoscedasticity

In case of heteroscedasticity, the variance of $u_i$conditional upon x_i is as follows:

$$E(u_i{}^2) = Var(u_i) = \sigma_i{}^2 \qquad \ldots (11.2)$$

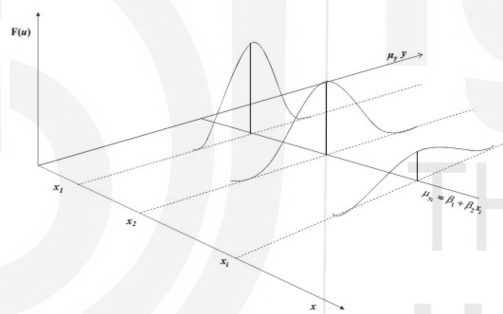

Fig. 11.2: Heteroscedasticity

11.3 CONSEQUENCES OF HETEROSCEDASTICITY

Heteroscedasticity has two important implications for estimation. First, the OLS estimators, while still being linear and unbiased, are no longer efficient. They no longer provide minimum variance estimators among the class of linear unbiased estimators (that is, they are not BLUE). Second, estimated variance of the OLS estimator is biased; so, the usual tests of statistical significance, such as *t* and *F* tests are no longer valid.

We prove these propositions below.

1. Unbiasedness of $\hat{\beta}$

Assume $y_i = \beta x_i + u_i$

Now, $\hat{\beta} = \frac{\sum x_i y_i}{\sum x_i{}^2} = \beta + \frac{\sum x_i u_i}{\sum x_i{}^2}$ $\qquad \ldots (11.3)$

Here, $E(u_i) = 0$

u_i are independent of x_i.

Thus, $E(\frac{\sum x_i u_i}{\sum x_i^2}) = 0$

Hence, $E(\hat{\beta}) = \beta$

That is, $\hat{\beta}$ is unbiased.

2. Efficiency of $\hat{\beta}$

To prove that the OLS estimator $\hat{\beta}$ is inefficient in the presence of heteroscedasticity, we show that the $Var(\beta^*)$ is less than $Var(\hat{\beta})$. β^* is the Weighted Least Square (WLS) estimator of β.

Now, $Var(\hat{\beta}) = E(\hat{\beta} - \beta)^2 = E(\frac{\sum x_i u_i}{\sum x_i^2})^2 = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$ … (11.4)

To find the WLS estimator of β, assume that, $\sigma_i^2 = \sigma^2 z_i^2$ and divide

$y_i = \beta x_i + u_i$ by z_i (here, z_i are known),

we get,

$\frac{y_i}{z_i} = \beta \frac{x_i}{z_i} + \frac{u_i}{z_i}$ … (11.5)

Let $\frac{u_i}{z_i} = v_i$

Now, the error term v_i has a constant variance σ^2. This method is called the Weighted Least Squares (WLS) wherein the i[th] term is weighed by $\frac{1}{z_i}$.

The WLS estimator $\beta^* = \frac{\sum\left(\frac{x_i}{z_i}\right)(y_i/z_i)}{\sum(\frac{x_i}{z_i})^2} = \beta + \frac{\sum\left(\frac{x_i}{z_i}\right)v_i}{\sum(\frac{x_i}{z_i})^2}$ … (11.6)

Since the expected value of the last term is zero, $E(\beta^*) = \beta$. Thus, the WLS estimator β^* is unbiased.

$Var(\beta^*) = \frac{\sigma^2}{\sum\left(\frac{x_i}{z_i}\right)^2}$ …(11.7)

and $Var(\hat{\beta}) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$

Substituting $\sigma_i^2 = \sigma^2 z_i^2$, we get,

$Var(\hat{\beta}_{HET}) = \frac{\sigma^2 \sum x_i^2 z_i^2}{(\sum x_i^2)^2}$ … (11.8)

Thus, $\frac{Var(\beta^*)}{Var(\hat{\beta})} = \frac{\sigma^2/\sum(\frac{x_i}{z_i})^2}{\sigma^2 \sum x_i^2 z_i^2/(\sum x_i^2)^2}$

On further elaboration,

$$\frac{Var(\beta^*)}{Var(\hat{\beta})} = \frac{(\sum x_i^2)^2}{\sum\left(\frac{x_i^2}{z_i^2}\right)\sum x_i^2 z_i^2} = \frac{(\sum(x_i z_i \frac{x_i}{z_i})^2}{\sum(x_i z_i)^2 \sum\left(\frac{x_i}{z_i}\right)^2} = \frac{(\sum a_i b_i)^2}{\sum a_i^2 \sum b_i^2} \quad \ldots (11.9)$$

where, $a_i = x_i z_i$ and $b_i = \frac{x_i}{z_i}$

The above expression is less than 1 implying that $Var(\beta^*)$ is less than $Var(\hat{\beta})$. Thus, the OLS estimator is inefficient in the presence of heteroscedasticity. The expression at (11.9) equals 1only if a_i and b_i are proportional to each other or ifz_i^2 is a constant in which case $\sigma_i^2 = \sigma^2 z_i^2$ shows that the error variances are equal.

Using $Var(\hat{\beta})$gives wider confidence intervals and standard errors are high. Hence, the *t*-ratios are smaller thus rendering a coefficient statistically insignificant even though it may found to be significant using WLS.

3. Var ($\hat{\beta}$) is biased

We cannot ignore the problem of heteroscedasticity because in the presence of heteroscedasticity, the variance of $\hat{\beta}$ is biased. Whether the bias is positive or negative depends on the relationship between σ_i^2 and x_i^2.

In the presence of heteroscedasticity,

$$Var(\hat{\beta}) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$$

We know from Unit 4 that, in the case of one explanatory variable,

$$Var(\hat{\beta}) = \frac{RSS}{n-1}\left(\frac{1}{\sum x_i^2}\right)$$

where RSS = Residual Sum of Squares

$$E(RSS) = E[\sum(y_i - \hat{y}_i)^2] = E[\sum(y_i - \hat{\beta}x_i)^2] = E[\sum(\beta x_i + u_i - \hat{\beta}x_i)^2]$$

Or, $E(RSS) = \sum \sigma_i^2 - \frac{\sum x_i^2 \sigma_i^2}{\sum x_i^2}$ … (11.10)

Hence, $E(Var(\hat{\beta})) = \frac{1}{(n-1)\sum x_i^2}[\sum \sigma_i^2 - \frac{\sum x_i^2 \sigma_i^2}{\sum x_i^2}]$

and,$Var(\hat{\beta}_{HET}) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$

Thus, $E(Var(\hat{\beta})) \neq Var(\hat{\beta}_{HET})$ … (11.11)

Therefore, the variance of $\hat{\beta}$ is biased. Since the standard errors of least square estimators are found to be biased, it invalidates the inferences drawn on the basis of the tests of significance.

Note that if $\sigma_i^2 = \sigma^2$ for all 'i', then $E(RSS) = (n-1)\sigma^2$. In that case, $E\left(Var(\hat{\beta})\right) = \frac{\sigma^2}{\sum x_i^2}$. Therefore, $Var(\hat{\beta}_{HET}) = \frac{\sigma^2}{\sum x_i^2}$. Thus, the bias disappears with homoscedastic error variances.

Check Your Progress 1

1) Consider the following equation where consumption of good X depends on the following: variables: $X = \beta_0 + \beta_1\, income + \beta_2\, price + \beta_3\, male + \beta_4\, education + u$

   $E(u \vdots income, price, male, education) = 0$

   $Var(u \vdots income, price, male, education) = \sigma^2 income^2$

   Transform this equation such that the error term is homoscedastic.

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2) "Heteroscedastic errors cause biased regression coefficients and biased standard errors". State whether True or False? Justify your answer.

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11.4 DETECTION OF HETEROSCEDASTICITY

In the case of heteroscedasticity, the OLS estimators are unbiased and consistent, but they are no longer efficient. This lack of efficiency makes the usual hypothesis testing procedure faulty and calls for remedial measures. Before we make any corrections for heteroscedasticity, we should first try to detect the presence of heteroscedasticity in the error terms. Two common methods are used to detect the presence of heteroscedasticity.

(i) We can plot the residuals against the predicted values of the dependent variable and examine the patterns of the residuals. If the variance of the residuals is not constant but increases or decreases with the independent variable, then there is a possibility of heteroscedasticity in the data set.

(ii) A more careful method is to break the sample into two or more sub-groups, on the basis of one of the independent variables (say, X), and then compute the error variance of each group. Our null hypothesis is that there is no difference among the variances of these groups. To test this hypothesis, we can use the chi-square test assuming that the error terms are normally and independently distributed. A problem with this method, however, is that the breakdown of these groups is rather arbitrary in terms of different ranges of X. Therefore, the test is viewed to be merely an indication of the presence of heteroscedasticity.

In practice, our data obtained from a sample survey, gives a single value of$y$corresponding to a value of X. Getting the entire range of Y population for chosen value of x is difficult. Hence, knowing ${\sigma_i}^2$ is also difficult. There are, however, other ways of detecting the presence of heteroscedasticity in the data.

1. Residual Plot

By plotting the estimated residuals$\hat{u}_i$ againstx_i, we can analyse whether there is a systematic pattern in the residuals which suggests heteroscedasticity in the errors. If the heteroscedastic error variance is found to be proportional to the explanatory variable (for example), it gives some prior knowledge which can be used to transform the data and correct for heteroscedasticity. Alternatively, ${\hat{u}_i}^2$ can be plotted against $\hat{y}_i$ and the residual pattern can be analysed. For instance, if a linear or quadratic pattern is observed, it indicates heteroscedasticity in the errors.

2. Auxiliary Regression

We can test for heteroscedasticity by (i) regressing $\hat{u}_i$ on ${x_i}^2$, ${x_i}^3$,... oron x_i, ${x_i}^2$, ${x_i}^3$...or $\hat{u}_i$ on ${\hat{y}_i}^2$, ${\hat{y}_i}^3$...., and (ii) testing if the regression coefficients obtained are statistically significant.

3. Park Test

In heteroscedasticity, the variance is different across the observations.

Let ${\sigma_i}^2 = \sigma^2 X_i^{\beta} e^{v_i}$ … (11.12)

By taking 'natural logarithm', we obtain, $ln\, {\sigma_i}^2 = ln\sigma^2 + \beta X_i + v_i$

If we take${\hat{u}_i}^2$as a proxy for${\sigma_i}^2$, we obtain

$ln\, {\hat{u}_i}^2 = \alpha + \beta X_i + v_i$ … (11.13)

Now, Park Test for detecting heteroscedasticity involves the following steps:

a) Run the original regression model despite the heteroscedasticity problem.

b) From the regression, obtain $\hat{u}_i$and square them, that is, ${\hat{u}_i}^2$. Then, find out $ln\, {\hat{u}_i}^2$

c) Run the auxiliary regression equation indicated in equation (11.13) using an explanatory variable present in the original regression model. Then run the regression against each X variable. In other words, we run the regression against $\hat{Y}_i$, the estimated value of Y_i.

d) Test the null hypothesis $\beta = 0$, i.e., there is no heteroscedasticity.

e) If the null hypothesis could not be rejected, we infer that there is no heteroscedasticity. In this case the value of the intercept can be accepted as the common, homoscedastic variance σ^2.

f) If the null hypothesis is rejected, then we infer that there is heteroscedasticity problem in the data set.

4. Glejser Test

The steps followed in the Glejser Test are similar to those of the Park Test. An important difference between both the tests is that in place of equation (11.13) we take different functional forms for the auxiliary regression equation.

The steps followed are:

a) Obtain the residual $\hat{u}_i$ from the original regression model.

b) Take absolute value $|\hat{u}_i|$ of the residuals.

c) Regress the absolute values of $|\hat{u}_i|$ on the X variable that is expected to be closely associated with heteroscedastic variance σ_i^2.

d) You can take various functional forms of X_i. Some of the functional forms suggested by Glejser are

$$|\hat{u}_i| = \alpha + \frac{\beta}{x_i} + v_i \qquad \dots (11.14)$$

$$|\hat{u}_i| = \alpha + \beta x_i + v_i \qquad \dots (11.15)$$

$$|\hat{u}_i| = \alpha + \beta \sqrt{x_i} + v_i \qquad \dots (11.16)$$

The above means that the Glejser test suggests various plausible (linear as well as non-linear) relationships between the residual term and the explanatory variable to investigate the presence of heteroscedasticity.

e) For each of the cases given, test the null hypothesis that there is no heteroscedasticity, i.e., $H_0: \beta = 0$ (no heteroscedasticity).

f) If H_0 is rejected, we conclude that there is evidence of heteroscedasticity.

You should note that the error term v_i can itself be heteroscedastic as well as serially correlated. For both the Park Test and the Glejser Test, the stochastic disturbance term v_i should satisfy the standard classical assumptions (see Unit 3).

5. Goldfeld-Quandt Test

The Goldfeld and Quandt (G-Q) Test is used if the sample size is not large. Firstl, the observations are ordered starting with the lowest value of ‘ X ’. The observations are divided into two groups such that the first group contains smaller values of the independent variable ‘X’ whereas the second group contains larger values of ‘x’. While segregating the two groups, some middle observations are omitted. Assuming that there are ‘n’ observations and ‘c’ observations are omitted, there remains$\frac{(n-c)}{2}$observations in each group. Separate OLS regressions are fitted in each group and the RSS are obtained corresponding to each group.

Each group’s RSS has$\left[\frac{(n-c)}{2}-k\right]$degrees of freedom. Here, k is the number of parameters (including the intercept term) that has to be estimated. Let RSS_2be the RSS from the group containing large values of ‘x’ while RSS_1 be the RSS from the group containing smaller values of ‘x’.

Now, assume $\lambda = \frac{RSS_2/df}{RSS_1/df}$ … (11.17)

If the error terms are assumed to be normally distributed and they are homoscedastic, then λ follows an F-distribution with $\left[\frac{(n-c)}{2}-k\right]$ degrees of freedom in both numerator and denominator. The null hypothesis (H_0) is the presence of homoscedasticity. The H_0 is rejected if the computed value of λ exceeds the critical F-value at the significance level assumed.

If the model has more than one independent variable, then any of the $x's$can be employed and the observations can be ranked according to it. Alternatively, the test can be conducted on each of the $x's.$ There are two limitations of the G-Q Test: (i) identification of the appropriate x variable to be used for ordering the observations. (ii) identifying the number of central observations that has to be omitted. Omitting a few central observations allows us to differentiate between the smaller variance group RSS_1 and the larger variance group RSS_2.

6. Breusch-Pagan Test

Two limitations of the Goldfeld and Quandt test mentioned above can be avoided by adopting the Breuschand Pagan (B-P) Test, which is an asymptotic test.

Let us assume that the variance of the error term is a function of certain variables such that,

$$\sigma_i^2 = f(\alpha_1 + \alpha_2 z_{2i} + \alpha_3 z_{3i} + \ ... + \alpha_m z_{mi}) \quad ... (11.18)$$

The steps to be followed are:

(i) Use the following regression model:

$$\sigma_i^2 = \alpha_1 + \alpha_2 z_{2i} + \alpha_3 z_{3i} + \ ... + \alpha_m z_{mi}$$

(iii) Test the null hypothesis that $\alpha_2 = \alpha_3 = \cdots = \alpha_m = 0$. If this null hypothesis is accepted, then σ_i^2 is homoscedastic. The function f(.) can take any form and the test is not affected by the functional form.

(iii) Let $\hat{\sigma}^2 = \sum \frac{\hat{u}_i^2}{n}$ ... (11.19)

(iv) Obtain S_0 = explained sum of squares from a regression of $\hat{u}_i^2$ on $z_2, z_3, ..., z_m$.

Then $\lambda = S_0/2\hat{\sigma}^4$ follows a χ^2 distribution with '*m*' degrees of freedom assuming normally distributed error terms and homoscedasticity.

7. White's Test

An assumption under the Breusch-Pagan Testis the normal distribution of the error terms. The Goldfeld-Quandt test, on the other hand, requires ordering of the observations. The White's test does not require the assumption that the error terms are normally distributed. Nor does it require ordering of observations. This test suggests regression of$\hat{u}_i^2$ on all the independent variables in the model as well as their squares and their cross products. In the case of three independent variables, for example, the White's test suggests regressing $\hat{u}_i^2$ on $x_1, \ x_2, \ x_3, \ x_1^2, \ x_2^2, \ x_3^2, \ x_1x_2, \ x_1x_3$ and x_2x_3 .

Let us take the case of two independent variables, x_2 and x_3. The steps to be followed are:

(i) Run the following auxiliary regression including the constant term:

$$\hat{u}_i^2 = \alpha_1 + \ \alpha_2 x_{2i} + \ \alpha_3 x_{3i} + \ \alpha_4 x_{2i}^2 + \alpha_5 x_{3i}^2 + \ \alpha_6 x_{2i} x_{3i} + \ v_i \quad ... (11.20)$$

(ii) Find the R^2 of the auxiliary regression equation given at equation (11.20).

(iii) For homoscedasticity, the null hypothesis is $H_0: \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = \alpha_6 = 0$

(iv) Consider that nR^2 asymptotically follows a χ^2 distribution with 5 degrees of freedom (n is the sample size). Here, the number of regressors without